

From 3D Documentation to Computer Vision: A Week in Brno

By Naira Khachatryan



Fig.1. A view of Brno's historic city centre, with the Moravian Museum and the Cathedral of St. Peter and Paul in the background. By Naira Khachatryan, 13 September 2026.

From 13 to 18 September 2026, I had the opportunity to take part in the ATRIUM Training School “Computer Vision in Archaeology”, hosted by the Institute of Archaeology of the Czech Academy of Sciences in Brno. I am currently writing my Master Thesis in Prehistoric Archaeology at Martin Luther University Halle-Wittenberg. My research focuses on Early Bronze Age ground stone tools from Shengavit, Armenia, using photogrammetry, structured-light scanning, 3D modelling and digital surface analysis. Since I already work with Python and digital archaeological data, my main motivation for attending the training school was to explore which new tendencies in computer vision and machine learning could be integrated into my existing research, particularly for analysing working surfaces and use-wear traces in Digital Environment.

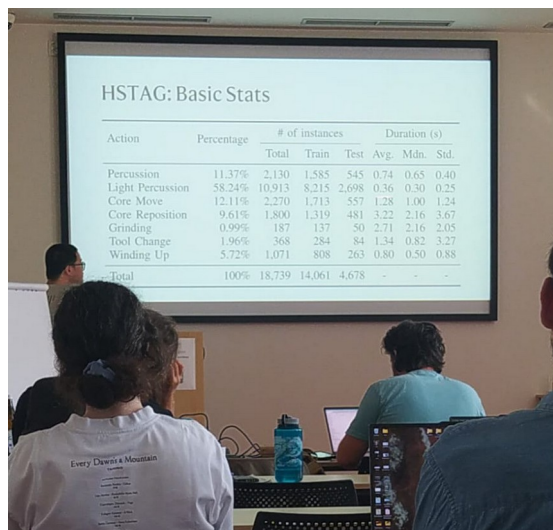
Different Backgrounds, Shared Interests



Fig.2. Getting to know each other's research projects and exchanging ideas during a collaborative group session at the ATRIUM Computer Vision in Archaeology Training School in Brno. By David Spáčil. 14 September 2026.

Monday began with introductions and group-work, and it was lovely to meet researchers at different stages of their academic careers, coming from a wide range of backgrounds and working on very different archaeological questions. What brought us together was a shared interest in archaeology, material culture and the possibilities of expanding our digital skills, particularly through machine learning. The first sessions introduced the fundamentals of computer vision and its applications in archaeology. We worked in groups with Python and Google Colab, and explored coding with AI agents. Since I already use Python in my research, particularly appreciated learning how to extend my existing workflows towards computer vision and translate archaeological observations into reproducible analytical steps. Working with coding agents, including the combination of Claude and Zed, was also a useful addition to my digital toolkit.

Tuesday: Annotation, Interpretation and Exchange



Action	Percentage	# of instances			Duration (s)		
		Total	Train	Test	Avg.	Min.	Std.
Percussion	11.37%	2,130	1,585	545	0.74	0.65	0.40
Light Percussion	58.24%	10,913	8,215	2,698	0.36	0.30	0.25
Core Move	12.11%	2,270	1,713	557	1.28	1.00	1.24
Core Reposition	9.61%	1,800	1,319	481	3.22	2.16	3.67
Grinding	0.99%	187	137	50	2.71	2.16	2.05
Tool Change	1.96%	368	284	84	1.34	0.82	3.27
Winding Up	5.72%	1,071	808	263	0.80	0.50	0.88
Total	100%	18,739	14,061	4,678	-	-	-

Figure 3. Cheng Liu presenting his Human Stone Toolmaking Action Grammar (HSTAG) project, which uses video annotation and machine learning to recognise and classify human actions involved in stone-tool production. By Naira Khachatryan, 15 September 2026.

Tuesday focused on archaeological artefact datasets and annotation. Using photographs from AMČR-PAS, we explored annotation formats such as COCO, YOLO and Pascal VOC, worked with CVAT and SAM 2, and discussed some familiar challenges in archaeological data: class imbalance, inconsistent vocabularies and missing metadata. We also had a chance to test our archaeological knowledge through ArchaeoTag, a guessing game that brought a welcome playful element into the session. For my own research, this was particularly relevant. I am already developing a qualitative basis for future quantitative analyses, including the consistent description of working surfaces, polish, striations and other use-wear traces. Seeing how other researchers approach category definitions, annotation decisions and uncertain features gave me valuable ideas for my own data. The afternoon was dedicated to participants' project presentations. It was nice to discover the range of research represented in the room, from palaeoenvironmental reconstruction and hyperspectral imaging to Palaeolithic art, stone carvings, drone imagery, stone-tool technology and 3D digitisation of seals. Hearing about colleagues' experiences and the different stages of their projects was a reminder that digital archaeology is a fast and constantly expanding field of possibilities.

After an intensive day, we exchanged our computers for a guided walk through Brno. It was a wonderful opportunity to learn more about the city's history, contemporary life and modern art. The evening continued around a pub table, where discussions moved naturally from archaeology and AI to their impact on society and many other topics. Sometimes the most interesting conversations happen after the official programme has ended.

Wednesday: From Microscopic Traces to Computer Vision

Wednesday was probably the day most directly connected to my current research. Working with microscopic use-wear images, we explored image preprocessing, contrast enhancement, masks, data augmentation and automatic segmentation using OpenCV. The practical exercises gave me useful tools. Techniques such as thresholding, edge detection and segmentation do not replace archaeological interpretation, but they can help transform qualitative observations into reproducible analytical procedures. I left the session with several ideas for my own research, particularly for connecting image-based observations with the surface geometry of my 3D models. During a break, we also had the opportunity to visit the institute's laboratories. I was impressed by the modern equipment and professional research environment. And, of course, even here scientific humour was not missing: Ea-nāšir, the ancient copper merchant who has somehow become an internet celebrity, had found his way into the archaeological world of Brno too!



Figure 5. Exploring the institute's modern laboratory facilities during a guided equipment demonstration. Institute of Archaeology, Brno. By David Spáčil. 16 September 2026.



And of course, a sense of humour was not missing here either! By Naira Khachatryan, 16 September 2026.

Thursday: Computer Vision Beyond Artefacts

Thursday took us to a completely different archaeological scale, with computer-vision applications in landscape archaeology, satellite imagery and rock art. It was exciting to see how broadly these methods can be applied and how creatively colleagues adapt machine learning to different research questions. Moving from microscopic use-wear to archaeological landscapes within the same week demonstrated that, although datasets and research questions vary considerably, many methodological challenges remain surprisingly similar.

Friday: Coins, AI and Looking Ahead

Friday morning introduced us to more advanced approaches using archaeological coin datasets. I particularly enjoyed learning how different computational methods can be combined to support more consistent identification and classification, reducing subjective variation while preserving the importance of specialist knowledge.

We also explored Hugging Face, an exciting platform for discovering AI and machine-learning models, datasets and ongoing developments. For someone interested in developing computational approaches to archaeological material, it is a valuable resource for finding inspiration, exploring existing methods, keeping up with new ideas and to share our expertise. The discussions about model interpretability, bias and reproducibility were equally important. A model may produce a convincing result without necessarily learning the archaeological characteristics we intended it to recognise. Understanding what happens between the input data and the final result therefore remains essential.

Thanks to our wonderful organisers, the week had a friendly and collaborative atmosphere from beginning to end. After a final lunch together, it was time to say goodbye to Brno and to the people who had made the experience so memorable.

What I Take Home from Brno

The most valuable outcome of the training school was gaining a clearer understanding of how computer vision and machine learning can complement my existing archaeological workflow.

In my Master's research, I am establishing a qualitative basis for the future quantitative investigation of ground stone tools. The training helped me think more critically about how this foundation can be structured to support computational approaches, from annotation and image processing to future comparisons with 3D surface data. I am returning with new knowledge, practical ideas for my research and a much broader perspective on the possibilities of machine learning in archaeology. Equally important are the conversations, new connections and warm memories of a week spent with people who share a passion for exploring the past through new methods.



Figure 6. A nice week of learning, exchanging ideas and making new connections. Group photo of the participants and organisers of the ATRIUM Computer Vision in Archaeology Training School, Brno, September 2026. By David Spáčil.(?)18 September 2026.

Many thanks to ATRIUM, the Institute of Archaeology of the Czech Academy of Sciences in Brno (ARÚB), MAIA, and especially the entire organising team for this wonderful opportunity, the inspiring programme and the welcoming atmosphere.